When Documents Don't Connect: Traceability Burden and Loopback Behavior in GMP Document Navigation

Sora Kim

Pharmaceutical Manufacturing Industry, South Korea

sorakim.lab@gmail.com

Abstract In pharmaceutical manufacturing environments governed by Good Manufacturing Practice (GMP) regulations, documents form interdependent networks, yet those relationships are typically represented as textual references rather than navigable structures. This paper argues that traceability is not only a property of documentation but an HCI problem about how document structure shapes navigation, and how navigation behavior reveals the structure practitioners are forced to work through. I introduce two design contributions that address this problem from complementary directions: an external traceability visibility prototype that makes document linkage navigable without modifying validated source files, and a behavioral navigation tracking prototype that operationalizes loopback behavior as an empirical indicator of document-structural friction. Two prototype-based studies with $N = 20$ GMP-experienced participants establish the empirical basis: Study 1 evaluated a graph-based Traceability Explorer over a synthetic corpus of 150 GMP records, identifying 31 missing CAPA links, 103 broken downstream links, and 21 lifecycle mismatch risks, with path reconstruction time reduced from approximately 7 minutes to approximately 4 minutes and all 20 participants completing target pathways successfully; Study 2 deployed a Cognitive Navigation Explorer during a standardized out-of-specification investigation scenario, observing loopback behavior across all 20 sessions (mean: approximately 5 loopbacks), with experienced practitioners averaging 2–3 loopbacks and less experienced participants averaging 7–10, while participants consistently attributed navigation difficulty to document structure rather than interface design. Three contributions follow: document traceability is introduced as a design problem specific to regulated environments; the no-modification constraint is identified as a novel design boundary for document navigation research; and making document relationships visible is argued to redistribute epistemic access by externalizing navigational structure that would otherwise remain implicit, unevenly distributed, and individually absorbed.

Keywords: document traceability, loopback behavior, GMP, pharmaceutical manufacturing, navigation friction, epistemic access, regulated environments, invisible work

Introduction

In pharmaceutical manufacturing environments governed by Good Manufacturing Practice (GMP) regulations, documents do not exist in isolation. A single deviation investigation may require navigating across standard operating procedures, analytical method protocols, validation records, change control documents, and specification files — each of which references others through textual cross-references rather than navigable links. When these references are complete and current, traceability is possible. When they are missing, outdated, or structurally ambiguous, the burden of reconstructing pathways falls entirely on the practitioner. The practitioner becomes, in effect, the missing infrastructure between documents.

This paper argues that traceability is not only a property of documentation — whether references exist, whether audit trails can be reconstructed — but a condition of interaction. When document relationships are only partially visible, practitioners must navigate through incomplete structures, and their navigation behavior reflects the structure they are working through. A loopback — a return to a previously visited document — is not simply evidence of inefficient searching. It is evidence that the path forward was not yet legible: that the document system did not resolve the judgment demand on the first pass. In information foraging terms (Pirolli and Card, 1999), a loopback is a rational response to weak information scent, not a navigational error.

I report two prototype-based studies in GMP environments. Study 1 develops and evaluates a Traceability Explorer — an external graph-based visibility layer over a synthetic corpus of 150 GMP records — and demonstrates that making linkage structure navigable reduces path reconstruction burden. Study 2 develops and evaluates a Cognitive Navigation Explorer that records document transitions, visit frequency, and loopback events during a standardized out-of-specification investigation scenario, demonstrating that loopback frequency varies systematically with experience in ways that reflect document structure rather than individual reasoning quality. Neither prototype modifies validated source documents — a constraint constitutive of GMP-regulated environments.

The contributions of this work are threefold. First, I introduce document traceability as an HCI problem specific to regulated environments, identifying the no-modification constraint as a novel design boundary for document navigation research. Second, I show that behavioral navigation tracking — loopback analysis in particular — can function as an empirical method for surfacing document-structural friction without deployment inside validated systems. Third, I offer a theoretical framing that connects document architecture to navigation behavior to accountability conditions: in GMP environments, whether a practitioner can reconstruct a defensible pathway depends not only on their expertise but on whether the document system makes its own structure legible.

Related Work

Documents as Organizational Action

The foundational premise of this study draws on Harper's (1998) ethnographic account of documentation practices at the IMF. Harper shows that documents are not neutral containers: their accessibility, their linkage to other documents, and the navigational demands they impose on users are constitutive of organizational action. When document relationships are visible and traversable, practitioners can reconstruct accountable pathways; when they are not, that reconstruction work falls to individuals.

Articulation Work and the Human as Missing Infrastructure

Schmidt and Bannon (1992) introduce the concept of articulation work to describe the coordinative labor that makes other work possible. In regulated document environments, cross-reference reconstruction is a form of articulation work: when linkage structure is incomplete or invisible, practitioners must manually bridge the gaps between documents, inferring relationships from partial cues and memory — labor that is not represented in any quality record.

Information Foraging and Loopback as Structural Signal

Pirolli and Card (1999) develop information foraging theory to explain how people navigate information environments: users follow information scent and move between information patches in ways that maximize gain relative to cost. A loopback is a rational response to weak or misleading scent. Applied to GMP document navigation, loopbacks indicate that the document system did not provide sufficient scent to support forward navigation, and that the practitioner was forced to reorient from a prior position.

Situated Action and Invisible Work

Suchman's (1987) account of situated action establishes that navigation through a document system is shaped by the immediate conditions of execution — what is visible, accessible, and interpretable at the moment a decision must be made. Star and Strauss (1999) show that work performed in the gaps of formal systems tends to become invisible. In regulated document environments, the manual reconstruction of broken traceability pathways is precisely this kind of invisible work.

Prototype-Based Evaluation

The methodological approach builds on Nielsen (1994)'s demonstration that think-aloud evaluation with small domain-specific samples reliably surfaces usability and friction patterns, and Hutchinson et al. (2003)'s logic of the technology probe: a prototype simultaneously as a design intervention and as an analytic instrument.

Method

Participants

Both studies were conducted with the same participant pool of $N = 20$, recruited from the same pharmaceutical quality team, with GMP work experience and direct familiarity with deviation or out-of-specification investigation workflows. Experience levels ranged from limited ($<$2 years, $n = 3$), intermediate ($n = 14$), to extensive ($\geq$5 years, $n = 3$). The same-sample design eliminates between-participant variance in domain knowledge as a confound.

Synthetic Document Corpora

Both studies used synthetic document corpora constructed in Python to reflect the structural and terminological characteristics of real GMP documentation without reproducing proprietary content — a domain constraint reflecting confidentiality obligations that preclude use of live quality records in external research.

Study 1: Traceability Explorer

The Study 1 corpus represented an interdependent network of 150 GMP records modeled as a directed graph using Python and NetworkX. Graph analysis identified three categories of linkage problem: 31 missing CAPA links, 103 broken downstream links, and 21 lifecycle mismatches.

Based on this analysis, I developed an external Traceability Explorer as a Streamlit web application organized into three tabs (Figures 1–3): a **Chain Viewer** presenting the four-record documentation chain for any selected record with missing and broken links flagged inline; a **Risk Summary** surfacing traceability risks at the dataset level; and a **Record Explorer** allowing filtering across the full corpus by record type, status, and product.

Participants completed three task-based traceability scenarios. Post-task measures included task completion time, path completion accuracy, document discovery rate, ease of use, and adoption intent.

Study 2: Cognitive Navigation Explorer

Participants navigated across seven synthetic GMP documents during a standardized out-of-specification investigation scenario. The Cognitive Navigation Explorer (Figures 4–6) recorded document-to-document transitions with action type (forward navigation, cross-reference jump, or loopback), visit count, time spent per document, and the full navigation sequence. A loopback was operationally defined as any return to a previously visited document where the intervening path had visited at least one other document.

Findings

Study 1: Traceability Visibility

Path reconstruction time reduced from approximately 7 minutes to approximately 4 minutes — a 30–60% reduction consistent across task types and experience levels. All 20 participants completed target pathway reconstruction, rated ease of use at 5.0 out of 5, and expressed adoption intent. Participants were particularly enthusiastic about the personnel handover use case: "every time someone leaves the team, the first three months of their replacement is basically archaeology." The prototype did not alter any underlying documents; it altered what relationships became immediately legible.

Study 2: Behavioral Navigation Tracking

Loopback behavior was observed across all 20 participant sessions, with a mean of approximately 5 loopbacks per session. The most common pattern was return to the ELISA Test Method SOP — not evidence of generic confusion but evidence that the first visit did not fully resolve the judgment demand. Experienced practitioners ($\geq$5 years) averaged 2–3 loopbacks; intermediate participants averaged 4–6; less experienced participants averaged 7–10 and were more likely to attribute difficulty to document structure than to interface design: "I kept going back to the ELISA SOP because I wasn't sure if I'd missed something — there were so many cross-references and I couldn't tell which ones actually mattered."

Discussion

Traceability as an Interaction Problem

Traceability in regulated environments is not only a documentation property but a condition of interaction. When document relationships are only partially visible, the burden of reconstructing

pathways is redistributed to practitioners, falling disproportionately on those with the least accumulated implicit knowledge. Making linkage structure visible changes who bears the cost of structural incompleteness.

Loopback as Method

Loopback behavior is an empirical indicator of document-structural friction rather than interface usability failure. Grounded in information foraging theory (Pirolli and Card, 1999), a loopback is a rational response to weak information scent. The mean of approximately 5 loopbacks per session and systematic variation by experience level suggest loopback frequency is a sensitive metric for document system evaluation — collectable through external prototypes without deployment inside validated systems.

Epistemic Access in Regulated Work

The experience asymmetry documented in Study 2 is not a training gap. It is a design gap. Experienced practitioners have internalized a map of the document system that the system itself does not provide. Making linkage structure visible externalizes that map — redistributing epistemic access rather than merely improving efficiency.

Limitations and Future Work

The use of synthetic corpora and a single-team participant pool are the primary limitations. Future work should deploy both prototypes against live document systems in production quality environments, particularly in personnel handover scenarios where the structural problem is well recognized and the intervention has no current analogue.

Conclusion

This paper introduced document traceability as a condition of interaction in regulated environments — a structural determinant of who can navigate defensible pathways, at what cost, and under what conditions. Study 1 demonstrated that a graph-based Traceability Explorer reduced path reconstruction time by 30–60%, with all 20 participants completing target pathways and expressing adoption intent. Study 2 demonstrated loopback behavior across all 20 sessions, with experience-stratified variation reflecting document structure rather than interface design. The experience asymmetry in loopback frequency is a design finding: the navigational support experienced practitioners have internalized through years of practice is not currently built into the document system itself.

References

Richard H. R. Harper. *Inside the IMF: An Ethnography of Documents, Technology, and Organizational Action*. Academic Press, 1998.

Hilary Hutchinson, Wendy Mackay, Bo Westerlund, Benjamin B. Bederson, Allison Druin, Catherine Plaisant, Michel Beaudouin-Lafon, Stéphane Conversy, Helen Evans, Heiko Hansen, Nicolas Roussel, and Björn Eiderbäck. Technology probes: Inspiring design for and with families. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI '03, pages 17–24. ACM, 2003.

Jakob Nielsen. *Usability Engineering*. Morgan Kaufmann, 1994.

Peter Pirolli and Stuart K. Card. Information foraging. *Psychological Review*, 106(4):643–675, 1999.

Kjeld Schmidt and Liam Bannon. Taking CSCW seriously: Supporting articulation work. *Computer Supported Cooperative Work*, 1(1–2):7–40, 1992.

Susan Leigh Star and Anselm Strauss. Layers of silence, arenas of voice: The ecology of visible and invisible work. *Computer Supported Cooperative Work*, 8(1–2):9–30, 1999.

Lucy A. Suchman. *Plans and Situated Actions: The Problem of Human-Machine Communication*. Cambridge University Press, 1987.

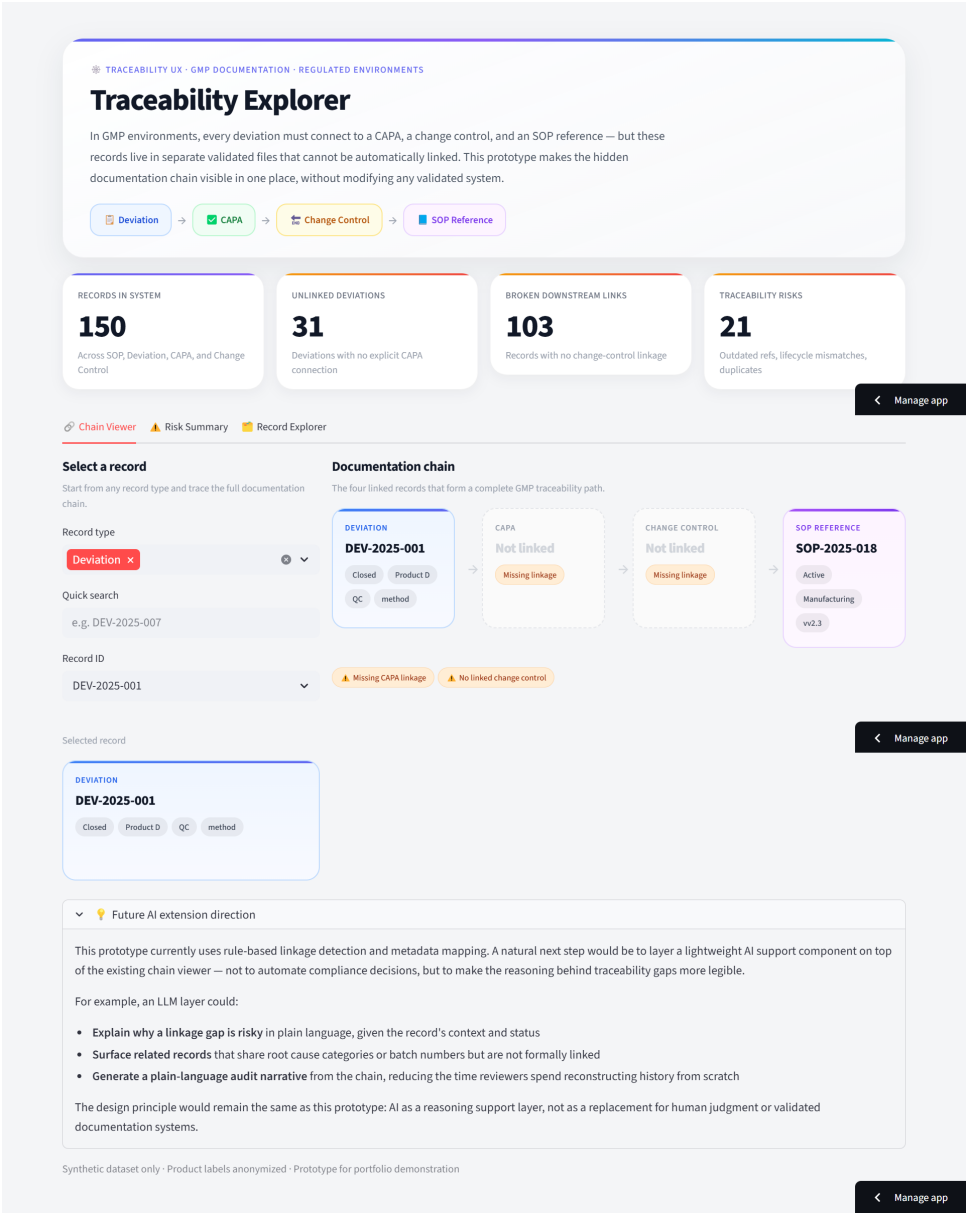


Figure 1: Traceability Explorer: Chain Viewer tab, showing the documentation chain for DEV-2025-001. CAPA and Change Control nodes are marked “Not linked,” while SOP-2025-018 is resolved as active and current.

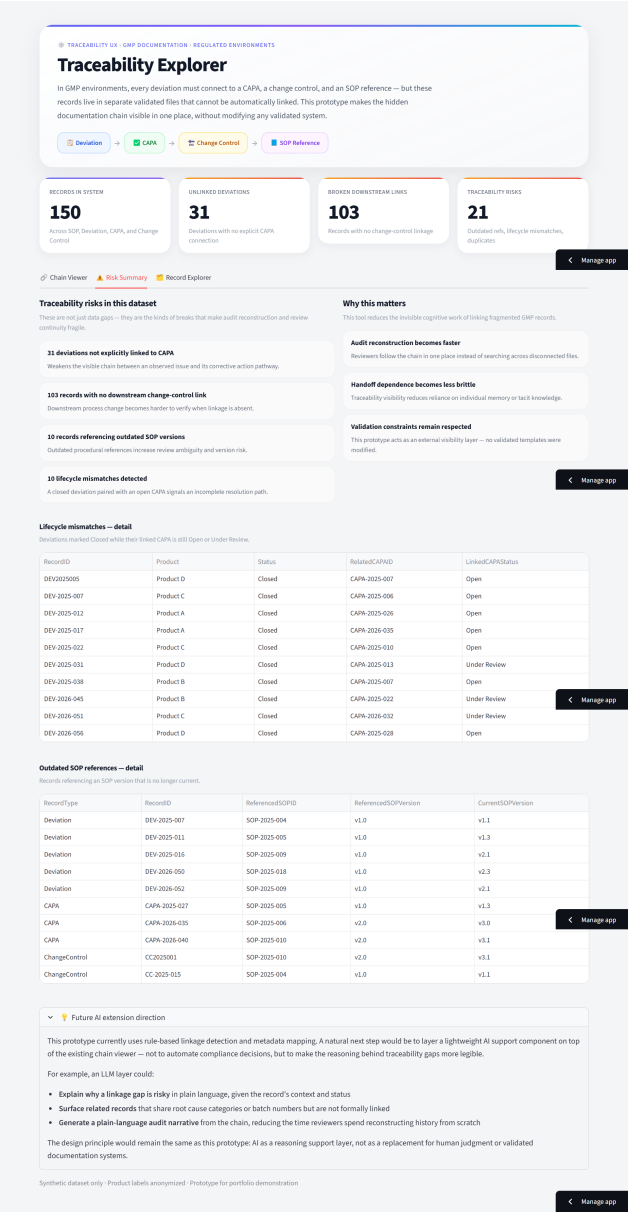


Figure 2: Traceability Explorer: Risk Summary tab, showing four risk categories. Lifecycle mismatches detail table shows 10 closed deviations whose linked CAPAs remain open. Outdated SOP references table shows 10 records referencing superseded versions.

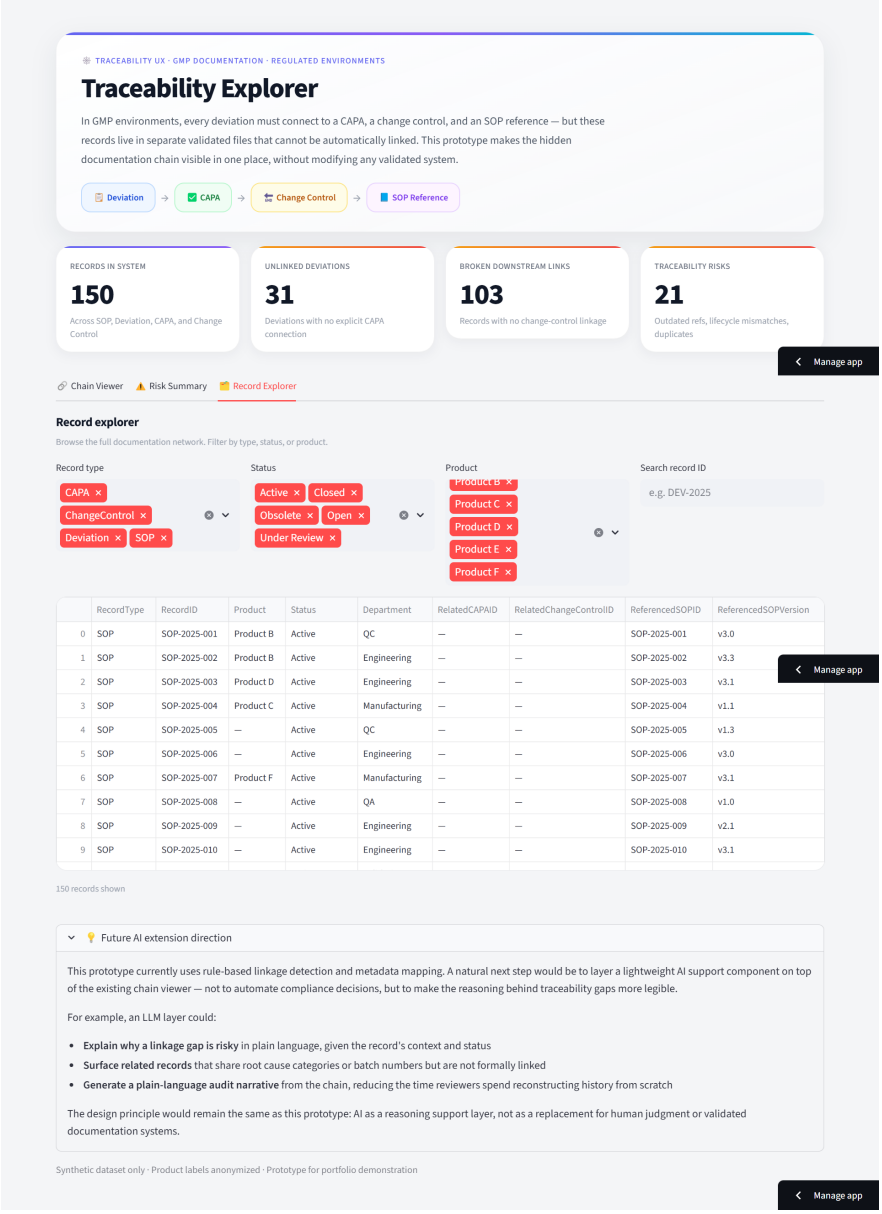


Figure 3: Traceability Explorer: Record Explorer tab, showing the full 150-record corpus with filters for record type, status, and product. Missing CAPA IDs and change control IDs are visible as empty cells.

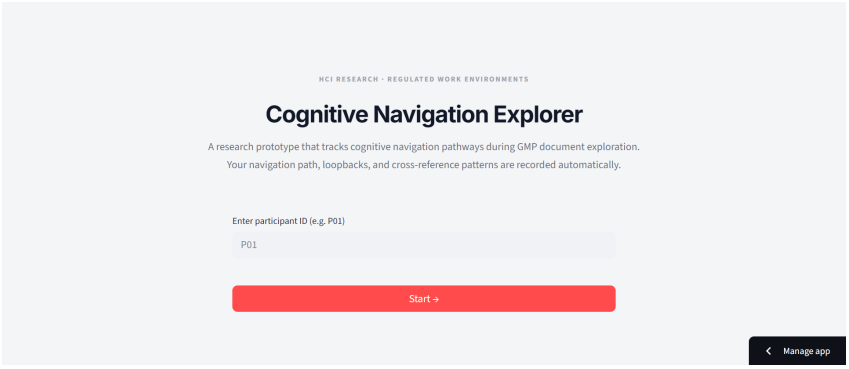


Figure 4: Cognitive Navigation Explorer: landing screen showing the study description and participant ID entry.

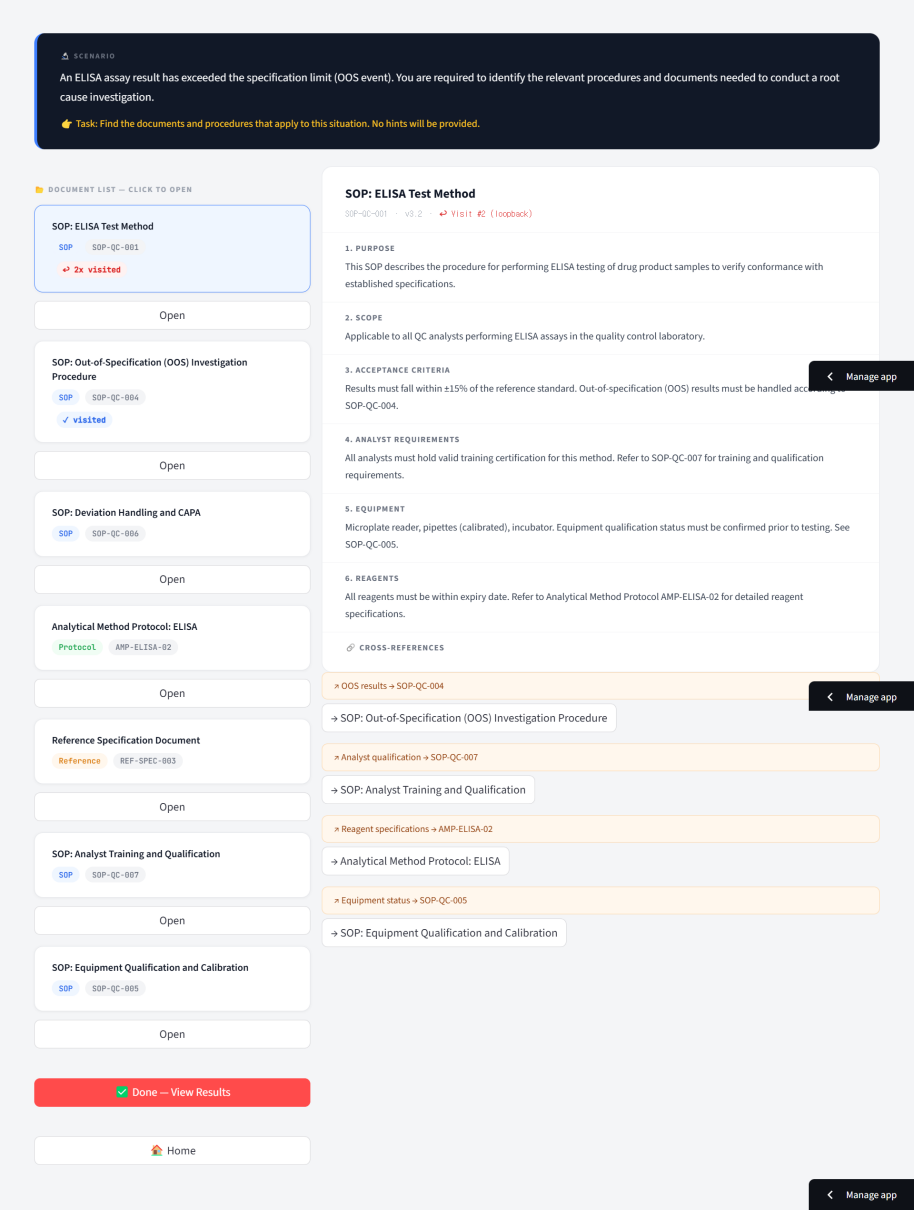


Figure 5: Cognitive Navigation Explorer: OOS investigation scenario in progress. The ELISA Test Method SOP is open at Visit #2 (loopback), indicated by the “2x visited” badge and the “Visit #2 (loopback)” label in the document header.

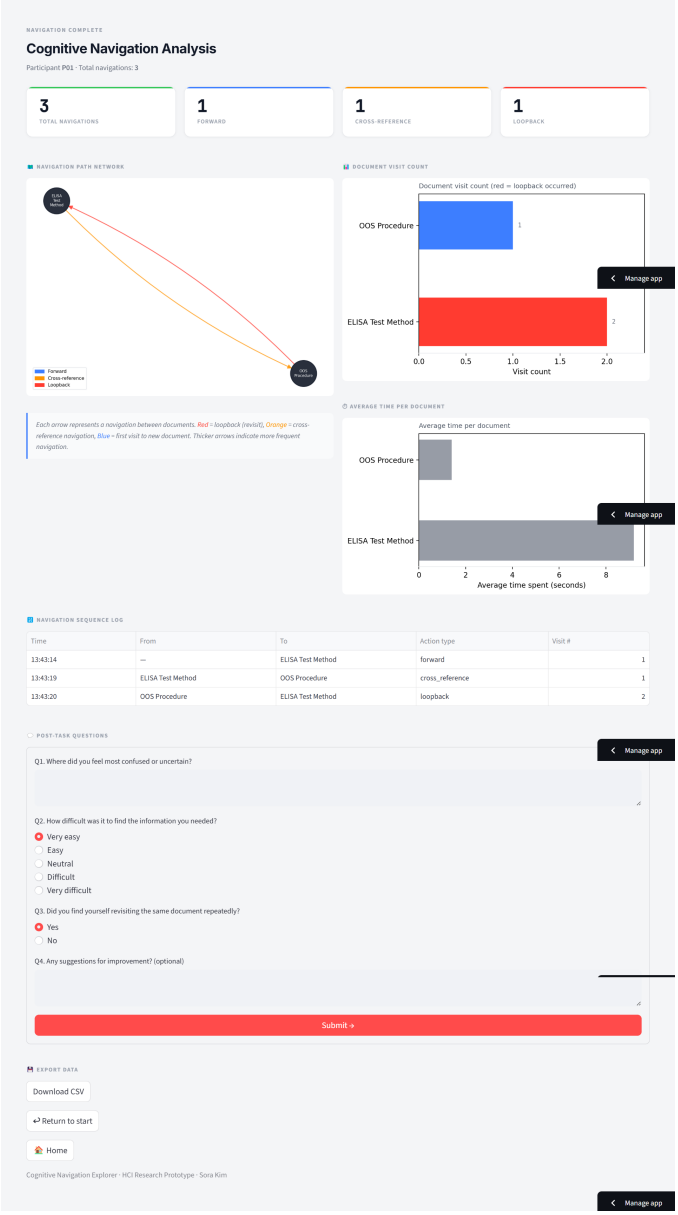


Figure 6: Cognitive Navigation Analysis: results dashboard for Participant P01, showing navigation path network and document visit counts. Red bars indicate loopback occurrence.